

or generic programs. They cannot process different types of data flowing into them. Instead they can do a small set of functions and are designed for only those. For Example, a household music system cannot run a fully functional web browser application but knows how to process FM Radio signals, ask the CD player module to read data, display track information on the LCD, process signals received from the infrared remote etc.

Digital Signal Processor (DSP)

A DSP is used to convert analog signals to digital which can be understood by microprocessors, microcontrollers and any other components in the system (SoC). DSPs are vast in their definition, implementation and use-cases - ranging from processing mobile signals, audio and video processing (encoding, decoding, converting, compressing, encrypting etc.) to reading pressures, temperatures and medical imaging. These signal processors run mathematical algorithms designed for the use case they serve.

There is an interesting side to DSPs when it comes to SoCs – many algorithms that DSPs execute on a hardware level can also be run on microprocessors at a software level – so you would not need another piece of hardware for decoding audio on your mobile; your quad core processor could run it. Why then are DSPs built as hardware chips? It's because most of these operations are compute-intensive mathematical operations. Running them on the generic microprocessor would waste energy, can get slow if the OS schedules the processor to do something else or because the algorithm requires parallel processing of multiple samples of data. Running a signal processing algorithms on the microprocessor can heat it up. This is why DSPs are separate from the main processor.

The list of use cases for DSPs is huge. DSPs are used in audio and video signal processing (sampling, encoding and compressing), speech recognition, reading natural environment (part of weather forecasting), seismology, medical imaging, radars and sonars (hence, scientific and military implementations) and a lot more. If there is a system which involves reading analog data and processing it in some way, a DSP is most probably part of the package.

For example, a missile targeting system might contain a GPS, an image processing system, a heat sensor, pressure sensor for calculating its position relative to its target. All those systems depend on a DSP to convert the information into digital format so that the main processor can run its algorithms to calculate the position and take further actions.